

ECE 311 – CAT 1

Network Analysis & Synthesis

Due date: 19th Jan, 2026 – 2359H, submission link will be provided

Answer ALL questions

Question 1

Find $v_C(t)$ for $t \geq 0$ in the circuit shown in Fig. Q1. The switch has been ON for a long time.

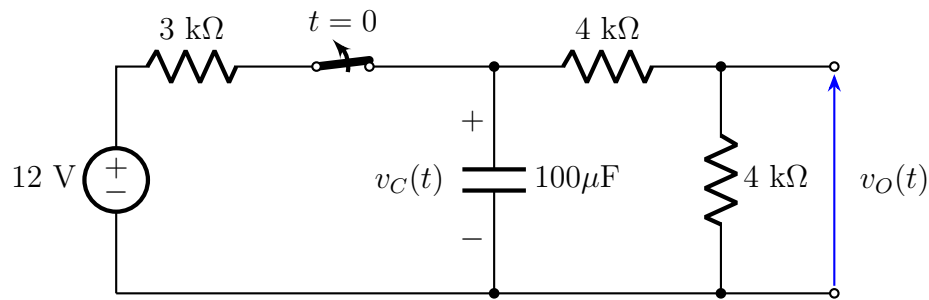


Fig. Q1

Question 2

Find $v_O(t)$ for $t \geq 0$ in Fig. Q2. The switch has been ON for a long time.

Question 3

The switch in the network in Fig. Q3 opens at $t = 0$. The switch has been ON for a long time. Find $i(t)$ for $t \geq 0$.

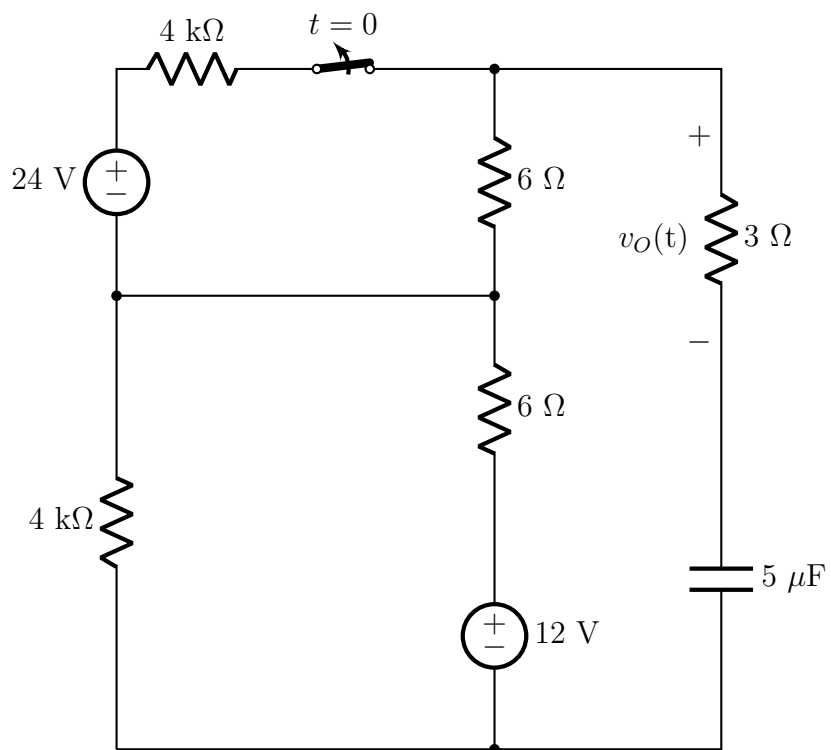


Fig. Q2

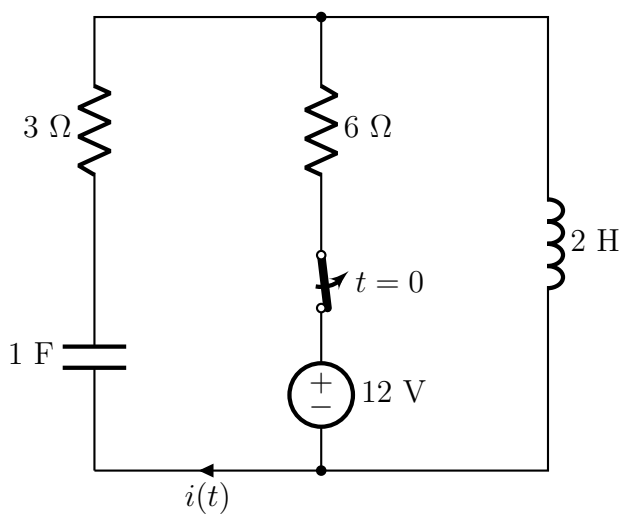


Fig. Q3